

# Venkata Naga Akash Ungarala

+1 (980) 337-8946 | [akash.ungarala@gmail.com](mailto:akash.ungarala@gmail.com) | [linkedin/akashungarala](https://www.linkedin.com/in/akashungarala) | [github/akashungarala](https://github.com/akashungarala) | Huntersville, NC

## Skills

|                                |                                                                                                                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>             | Typescript, Python, SQL, C Sharp                                                                                                                                                                                                          |
| <b>Backend Development</b>     | Node.js, Express.js, Next.js, Flask, FastAPI, Django                                                                                                                                                                                      |
| <b>Async Processing</b>        | Celery (Jobs, Promises), asyncio, RabbitMQ, event-driven architecture                                                                                                                                                                     |
| <b>Testing</b>                 | Jest, PyTest, Postman (APIs), Swagger (request/response validation with YAML)                                                                                                                                                             |
| <b>Frontend</b>                | HTML, CSS, Bootstrap, Javascript, React                                                                                                                                                                                                   |
| <b>Databases &amp; Storage</b> | Postgres, MySQL, MongoDB, SQLite, GCP (Cloud Storage, Cloud SQL, Spanner, BigQuery, Firestore, Firebase), AWS (RDS, DynamoDB, S3)                                                                                                         |
| <b>Database ORMs</b>           | SQLAlchemy, Django ORM, TypeORM                                                                                                                                                                                                           |
| <b>Data Formats</b>            | JSON, XML, YAML, HTML, CSV, Protocol Buffers                                                                                                                                                                                              |
| <b>GCP Ecosystem</b>           | Pub/Sub, Cloud SDK, Cloud Functions, Cloud Operations (Stackdriver), Kubernetes Engine (GKE), IAM, Cloud KMS, Container Registry (GCR), Maps Platform, Dialogflow, Cloud Storage (GCS), Cloud SQL, Spanner, BigQuery, Firestore, Firebase |
| <b>AWS Ecosystem</b>           | EC2, Lambda, SNS, API Gateway, Cloud Watch, KMS, IAM, RDS, DynamoDB, S3, AMG, AMP                                                                                                                                                         |
| <b>Version Control</b>         | Git, GitHub, GitLab, Bitbucket                                                                                                                                                                                                            |
| <b>CI/CD &amp; DevOps</b>      | Bitbucket, GitLab Pipelines, Jenkins, Spinnaker, Docker, Kubernetes, Prometheus, Grafana, Helm                                                                                                                                            |
| <b>Other Skills</b>            | Open API V3, Auth0, Apache Airflow, Agile Scrum, SDLC, Microservices, Distributed Systems, REST                                                                                                                                           |

## Experience

### Kforce Inc, NC – Senior Software Developer (Client: Fox, AZ - Sr. Back End Developer)

Sep 2024 – Dec 2025

- Spearheading the design and enhancement of CaptionExchange, a scalable real-time live captioning service supporting both AI/ASR and human-generated captions with robust routing and failover mechanisms.
- Developed and maintained backend services on a Kubernetes-based Service Platform, integrating Dockerized containers, AI/ASR providers, and dynamic routing infrastructure.
- Implemented scalable and resilient Agent and Gateway services to manage decoded audio transport, caption injection via edge devices, and human captioner session control.
- Built APIs and tooling to orchestrate broker, agent, and caption routing logic, supporting real-time buffer management and automatic failover for uninterrupted caption delivery.
- Instrumented logging and monitoring solutions for Docker-based agent servers; designed debugging workflows using server logs, NMOS state files, and real-time UI status panels.
- Led operational support and configuration for SDVN environments, managing server authentication via token-based identities and integrating with NMOS registries.
- Collaborated cross-functionally across media and broadcast teams to support seamless switching between AI and human captioning, ensuring accurate visibility in the CaptionExchange UI.
- Developed test frameworks with Jest, ensuring comprehensive unit and integration test coverage; enforced CI pipelines for consistent quality across environments.
- Introduced structured Winston logging and environment-specific configurations across test, dev, and prod environments to streamline observability and debugging.
- Containerized services with Docker and managed secure deployments through internal jumpbox access, Kubernetes clusters, and Helm charts.
- Enhanced developer productivity through Biome-based formatting/linting and modular TypeScript project configurations.
- Championed API-first design principles and maintained OpenAPI documentation for seamless backend-frontend integration and external consumer adoption.

### Kforce Inc, NC – Advanced Software Developer (Client: Fidelity, NC - Software Engineer)

Oct 2023 – Jul 2024

- Led the design and development of a custom SLO Tracker solution integrated with the Open Telemetry gateway stack and AWS backends, emphasizing tool reuse and an API-first approach for scalability.
- Enabled teams to define SLOs using an alerts-as-code model, set error budgets, and create SLOs directly from Alert Definitions.
- Developed the backend with Python and FastAPI, using AWS Lambda, Terraform, and API Gateway for deployment, and PostgreSQL RDS with Alembic for data management.
- Integrated SNS triggers for automated SLO alerts and proactive breach notifications.
- Developed a responsive and efficient ReactJS application with TypeScript for the SLO solution's frontend. The interface provided centralized visibility into SLOs, targets, and deficits.

- Leveraged cloud-native serverless and container solutions, ensuring asynchronous communication and seamless integration with the observability pipeline.
- Implemented a proof of concept for high cardinality metrics using Amazon Managed Service for Prometheus and developed a Grafana dashboard for visualizing metric cardinality details to identify opportunities for optimization.
- Ensured code quality and reliability through comprehensive automated testing using Jest for Node.js and PyTest for Python applications.

#### **Niantic Inc, CA – Software Engineer II**

**Nov 2021 – Aug 2023**

- Designed and implemented scalable, contextual location-based ad solutions, serving millions of users daily and providing real-time performance tracking, accurate user segmentation, and comprehensive analytics to deliver measurable value to brand partners.
- Developed a Node.js application with Express.js, TypeScript, protocol buffers, and Google Cloud Spanner, optimizing data storage and retrieval with PostgreSQL and complex queries using TypeORM.
- Led test-driven development (TDD) and implemented parameterized unit tests using Jest for Node.js microservices and PyTest for Flask microservices, ensuring high code quality and reliability.
- Developed and maintained high-performance RESTful APIs using TypeScript (Node.js) and Python (FastAPI), facilitating real-time data access and enhancing system responsiveness.
- Implemented object-oriented programming (OOP) principles for maintainable and extensible codebases, and developed ETL processes using Python/TypeScript and relational databases.
- Developed and maintained ETL (extract, transform, and load) processes using Python/Typescript and relational databases.
- Implemented Celery for asynchronous task processing and job scheduling, reducing system latency and improving overall system responsiveness.
- Orchestrated OpenAPI specifications and Swagger documentation for clear, up-to-date API documentation along with the ability to validate requests and responses of the APIs.
- Containerized applications with Docker, managed Kubernetes deployments using Helm charts, and streamlined CI/CD processes with GitLab Pipelines, reducing software defects by 40%.
- Contributed to the Pokemon Go game by developing the telemetry mechanism for WebAR ads and establishing tiered reward patterns, laying the foundation for future reward flows in the Ads SDK.
- Crafted server-side AdTech solutions for planet-scale game servers, platform services, large data stores, secure networking, and real-time analytics, supporting Niantic's products.
- Led the breakdown of complex problems into actionable user stories, optimized team efforts to meet release timelines, and advocated for agile best practices to deliver solutions despite specification ambiguities.
- Mentored engineering teams, promoted coding standards, and instilled best practices in design and testing, fostering a product-focused mindset and improving overall code quality.
- Spearheaded user-centered testing methodologies, covering comprehensive scenarios and edge cases, enhancing feature quality and elevating customer satisfaction.

#### **Adara Inc, CA – Senior Backend Engineer**

**Feb 2019 – Nov 2021**

- Developed and improved data pipelines to better support needs from data science, business applications and reporting.
- Spearheaded the development of a real-time analytics dashboard using Flask, and SQLAlchemy, providing stakeholders with critical insights into data trends and KPIs.
- Implemented fine-grained access control and role-based authentication systems in Python applications, enhancing data security and privacy for users and organizations.
- Designed and implemented a custom ORM layer on top of Postgres, optimizing database schema for complex queries and reducing query times by 30%.
- Utilized Google BigQuery for data warehousing and analysis, enabling data-driven decision-making through SQL.
- Engineered a high-throughput data ingestion system in Python using Cloud Pub/Sub, handling terabytes of data daily.
- Leveraged Cloud Pub/Sub for event-driven microservices communication for seamless service communication, decoupling services and achieving better scalability and maintainability.
- Enhanced system responsiveness with Celery for asynchronous task processing and job scheduling.
- Orchestrated complex workflows with Directed Acyclic Graphs (DAGs) using Apache Airflow.
- Developed templated framework (boilerplate code) as python packages so that any engineer can use them to spin-up a new flask application and start implementing APIs as needed, and to deploy the app in quick time, just make necessary changes to the default YAML config file.
- Advocated for Python packaging practices to enhance code reusability and maintainability.
- Contributed to Adara's core engineering infrastructure by implementing automated CI/CD setup process (YAML based configuration), developing design/code standards, tools, dashboards, alerts, reporting, documenting best practices and tooling decisions, investigating opportunities for cost optimizations (GCP and any other external tools) across engineering.

- Developed build and deployment automation using Bitbucket pipelines to run all the tests and prevent regressions.
- Leveraged Spinnaker for automated, safe, and efficient deployments.
- Maintained real-time visibility into application performance using Cloud Operations (Stackdriver) for logging, Prometheus for monitoring and alerting, Grafana for dashboards.
- Containerized applications using Docker and orchestrated deployments with Kubernetes, resulting in improved scalability, resource utilization, and zero-downtime updates.
- Integrated Auth0 for OAuth2-based secure user authentication and Firebase for user management, implementing role-based access control (RBAC) and fine-grained permissions to enhance application security.
- Architect breakdown of large problems to be solved, features as per the release timelines into user stories and plan them for upcoming sprints along with the bugs/stories based on the priority, dependencies and needs of other teams, product.
- Build robust, scalable and low-latency systems, tools and APIs for internal/external teams and prototype new products and platforms combining real time data with machine learning.
- Manage complex technical integrations with customers and partners including building new APIs and integrating with 3rd party marketing platforms.
- Mentor, train and guide engineers in adopting product thinking, coding standards, design and testing patterns. Recommend user perspective testing approaches that can cover all the scenarios including edge cases considering how the feature should behave or what would the customer want next.

**LTI, CA – Optional Practical Trainee (Client: Apple Inc, CA - Senior Software Developer)**

**Dec 2018 – Feb 2019**

- Developed key features for a Learning and Management Services (LMS) application used by Apple retail employees to undergo training recommended by their managers.
- Designed and developed RESTful APIs using FastAPI with Swagger, significantly enhancing data accessibility and providing clear and comprehensive API documentation.
- Integrated these APIs into client applications for iOS, iPad, and macOS, ensuring seamless communication between the backend and frontend.
- Implemented GitLab Pipelines for CI/CD, streamlining software delivery with automated testing and code quality checks, resulting in faster and more reliable releases.

**BCforward, CA – Action Support Engineer (Client: Google, CA - Action Support Engineer)**

**Oct 2017 – Dec 2018**

- Supported the product Dialogflow, which incorporates Google's machine learning expertise and Google Cloud Speech-to-Text, powering actions for over 400M+ Google Assistant devices.
- Developed Dialogflow chatbot prototypes in Python, Node.js, and Java using client libraries, and created demo apps for iOS and Android by integrating Dialogflow REST APIs. These chatbot apps processed natural language requests and returned real-time dynamic responses from external APIs and backend databases via webhook setup.
- Assisted developers by debugging and troubleshooting issues with APIs and SDKs, ensuring smooth integration and functionality.
- Tested APIs across all versions and product integrations with platforms like Google Assistant, Facebook, Twitter, Slack, and Skype during the build and release phases.
- Recommended scalable agent designs to enterprise partners, providing prototypes that best fit their AI scenarios.

**ReelBabble LLC, CA – Mobile Application Developer**

**Apr 2017 – Jul 2017**

- Developed server-side features for ReelBabble, a video-based movie reviews iOS app ([App Store](#)) following scrum agile and involved research, brainstorming, visual design, prototyping, development, user testing, and project planning.
- Implemented an AuthService to communicate with Firebase for user management, real-time notifications, and real-time data synchronization, resulting in a 20% increase in user retention.
- Designed and implemented the features like the sign-up process, email, social login and consumed the external API ([themoviedb.org](#)) to render content (movie details, posters, trailer/teaser URLs, etc.) in the application.
- Developed dynamic custom UI from visual mock-ups for the mobile app and also server-side functionality to implement features like browse, upload, like, dislike video reviews by connecting to AWS S3 for upload and retrieval of video content.
- Integrated S3 with AWS Lambda functions to automate file processing tasks, such as image resizing, log analysis, and data transformation.
- Leveraged S3 as a scalable and reliable storage backend for web applications, improving content delivery and user experience.

**UNC Charlotte, NC – Graduate Research Assistant**

**May 2016 – Dec 2016**

- Designed and developed features for NatureNet, a citizen science project enabling users to share location-based nature observations and contribute design ideas and challenges across web, Android, and iOS platforms.
- Implemented a location-based service platform using Python Flask within the Google Cloud Platform (GCP) ecosystem, enhancing user engagement and location accuracy through seamless frontend integration.

- Utilized Google Firebase for user authentication, Firestore for storing user and observation details, and Cloud Storage for image uploads. Employed asynchronous data fetching techniques to optimize content rendering in the application.
- Consumed RESTful APIs exposed by the Python Flask application in web, Android, and iOS apps, adhering to visual mockups finalized during sprint meetings.

#### **Infosys Ltd, India – Systems Engineer**

**Feb 2014 – Jul 2015**

- Developed features for the web portal implemented in Python Django, an innovative & self-service ecosystem for the client PG&E (Pacific Gas and Electric Company) to improve the user experience of the online portal for managers to perform actions related to the company and employees.
- Implemented features such as sign-up, login modules, store, retrieve data related to organization and employees by connecting the web application to a MySQL database and consuming third party REST APIs as per the workflow.
- Hands-on experience in writing SQL queries to perform CRUD (Create, Read, Update, Delete) operations on the data and work with tables, stored procedures, functions, triggers, views, indexes and constraints in MySQL database.

#### **Education**

|                                                  |                                                   |                     |
|--------------------------------------------------|---------------------------------------------------|---------------------|
| University of North Carolina at Charlotte, USA   | Master of Science in Computer Science             | Aug 2015 – Dec 2016 |
| Jawaharlal Nehru Technological University, India | Bachelor of Technology in Electronics Engineering | Sep 2009 – May 2013 |